

Nicolas Alvarez Arce

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EDUCATION

Loyola Marymount University

Los Angeles, CA

Bachelor of Science in Mechanical Engineering, Minor in Math

May 2025

- **Relevant Coursework:** Computer Aided Design (CAD), Algorithms & Applications (MATLAB & Python), Heat Transfer, Thermodynamics, Finite Element Analysis, Rapid Prototyping, Robotics, Aerodynamic.

EXPERIENCE

Boeing

Los Angeles, CA

Student Research Assistant at the Materials Lab

November 2022 - May 2025

- Performed hardness and tensile testing on 30+ AlSi10Mg 3D-printed samples to measure strength loss from overaging, providing Boeing with data used to refine post-processing heat-treatment standards.
- Operated optical microscopy and SEM to analyze microstructures of 30+ AlSi10Mg samples, correlating findings with tensile and hardness results to assess processing effects on mechanical performance.
- Partnered with Boeing engineers to align research objectives with industry needs; presented findings at ASME meetings and to 20+ professionals, contributing to a manuscript currently under review for journal publication.

LMU Fluids of Astrophysical Bodies (FAB) Laboratory

Los Angeles, California

Mechanical Engineer

August 2023 - December 2023

- Developed a Rotary Convection table to investigate thermal convection on icy moons, involving (CAD) modeling and Finite Element Analysis (FEA) for structural integrity and performance optimization.
- Designed a cryogenic biaxial “expanding-box” apparatus to study ice creep under simulated icy-moon conditions, enabling quantitative strain-rate measurements (10^{-8} – 10^{-3} s $^{-1}$) across variable stress (0.1–10 MPa) and temperature (90–273 K) ranges for a collaborative proposal to NASA JPL

LMU Thermal Fluids Lab

Los Angeles, California

Student Research Assistant

May 2023 - August 2023

- Researched microdroplet lifetime on heated surfaces to improve cooling performance for electronic PCB and vapor chamber applications.
- Designed and fabricated experimental hardware (CAD, machining, CNC, thermocouple-embedded copper surface) and conducted droplet impact experiments using multiple liquids.
- Utilized DAQ systems (LabVIEW) and MATLAB to collect, process, and analyze sensor data, determining droplet residence times and heat-transfer behavior.

Ruhrpumpen

Monterrey, Mexico

Project Engineer Intern

May 2024 - October 2024

- Managed purchase orders, inspection plans, and supplier documentation while communicating with clients and suppliers to ensure on-time delivery and compliance with API, ASME, and ISO standards.
- Performed process mapping, time studies, and KPI/defect analysis to identify bottlenecks and cut cycle times by up to 15%, improving overall efficiency.
- Partnered with manufacturing and product engineering teams to implement Lean/Six Sigma improvements, enhancing product quality, reducing rework, and maintaining alignment with client specifications.

SKILLS

Spanish (Native) • SolidWorks • Abaqus • GD&T (ASME Y14.5) • MATLAB • Python • C/C++ • Arduino • Machining • Prototyping • 3D Printing (FDM/SLA) • Lean Six Sigma Green Belt • Google Data Analytics • Prompt Engineering & AI Workflows (Claude, Gemini).

Projects: Rubik’s Robot (CV/Mechatronics); 6-DOF Stewart Platform (Motion Control); IGVC Capstone (Autonomous Nav); Two-Stage Rocket (Propulsion); Kitesurfing Gear (CAD/Action Cam Design); Mechanical Dictamens (Equipment Failure).

Leadership/Activities: President, LMARS (Rocketry); Treasurer, Climbing Club (300+ members); D1 Men’s Crew (Varsity); Rock Climbing, Photography, & Outdoors.